

Is a Pharmacy Degree Worth It in the AI Era?

The complete profile — the full scores by track, the six factors behind them, the honest downsides, and what to ask before you commit.

4.9

AI EXPOSURE · COMMUNITY PHARMACIST

where 10 is most at risk

Read this one first; send the student version to them.



Before you read this — a note for parents

You're likely reading this before your child is. That's normal, and it's worth pausing on how you use it.

Don't lead with the scores. If you open with a risk number, they either get frightened and shut down, or defensive about a decision they've already made.

What this is. A facts document, not a recommendation.

A practical note: there's a separate short version written directly to them. Send them that one.

And one thing specific to pharmacy. This is the profile most likely to surprise a family who assumed healthcare is uniformly protected. Pharmacy is a licensed clinical profession requiring a doctorate — and it scores worse than nursing, medicine, dentistry, veterinary medicine and physical therapy.

The reason is specific and worth understanding rather than alarming: it's about *what the licence reserves*, and section 4 explains it. There is also a genuine reversal happening in the job market that most careers advice hasn't caught up with.

The short answer

Better than the average career, and the weakest position in licensed healthcare.

Clinical and hospital pharmacy scores **4.0** out of 10. Community and retail pharmacy scores **4.9**. Central fill and mail-order scores **6.9**.

Pharmacy is the clearest demonstration in this index that a licence is only as protective as the act it reserves — and pharmacy's licence reserves dispensing, which is exactly the part a machine does well.

The good news is that the profession is moving toward the protected end faster than most people realise, and the job market is reversing in a way that hasn't reached most careers advice.

The scores

TRACK	2023	2025	FALL 2026	3-YR MOVEMENT	BAND
Clinical / hospital pharmacy	3.4	3.7	4.0	+0.6	Low–Moderate
Ambulatory & specialty pharmacy	3.8	4.1	4.4	+0.6	Moderate
Community / retail pharmacy	4.2	4.6	4.9	+0.7	Moderate
Pharmacy technician	4.8	5.3	5.7	+0.9	Moderate
Central fill / mail order	5.8	6.4	6.9	+1.1	Moderate–High

Scores run 1–10, where 10 is most at risk. For context: the median profession in this edition sits around 5.5, bedside nursing scores 2.8, dentistry 2.7, physical therapy 2.5, and entry-level software development 8.1.

Read the table this way: every track in pharmacy scores worse than every clinical track in nursing, dentistry or physical therapy. That isn't an accident of measurement — it follows directly from what the work consists of.

Why pharmacy scores worse than other licensed healthcare — the six factors

Here's how community pharmacy answers them.

HOW MUCH OF THIS JOB CAN AI ALREADY DO? (35% OF THE SCORE)

A substantial share — and considerably more than in other clinical professions.

Prescription verification against interaction and dosing rules. Inventory management and ordering. Insurance adjudication and claims. Refill authorisation. Dispensing itself, through automated counting and packaging. Patient information leaflets and standard counselling material.

Dispensing is a rules-based, structured, high-volume task with a clear correct answer. That is close to a definition of automatable work.

What resists: clinical judgment on a complex regimen, catching the error nobody's system flagged, counselling a frightened patient about a diagnosis they've just received, and the professional accountability that attaches to a signature.

HOW HARD WILL IT BE TO LAND THAT FIRST JOB? (15%)

Genuinely contested, and section 5 covers a reversal most careers advice hasn't registered.

DOES IT HAVE TO BE DONE IN PERSON, WITH YOUR HANDS? (15%)

Far less than other clinical professions, and this is the crux.

A nurse must be physically present to turn a patient. A dentist must be physically present to work in a mouth. A pharmacist verifying a prescription does not need to be in the same building — which is why central fill and mail-order pharmacy exist at all, and why they score 6.9.

DOES SOMEONE NEED A HUMAN THEY CAN TRUST AND HOLD RESPONSIBLE? (15%)

At the clinical end substantially; at the dispensing end less so.

A hospital pharmacist advising on a complex regimen carries real accountability. A pharmacist verifying a routine repeat prescription is performing a check that a system can increasingly perform.

DOES THE LAW REQUIRE A LICENSED HUMAN? (10%)

Comprehensively — and section 4 explains why this protects less than it appears to.

HOW OFTEN DOES THE JOB HIT GENUINELY NEW, HIGH-STAKES SITUATIONS? (10%)

At the clinical end frequently; in community dispensing rarely. Most prescriptions are routine by design. That's the system working correctly, and it's also what makes the work automatable.

A licence is only as protective as the act it reserves

This is the most important idea in this profile, and it completes an argument that runs through several others in this index.

Licensing is normally the most durable protection we score. It moves at the speed of legislation rather than technology, and no advance in capability repeals a scope-of-practice rule. Pharmacy has it comprehensively: dispensing prescription medication requires a licensed pharmacist, full stop.

So why does pharmacy score worse than nursing?

Because a licence protects a specific act — and the acts differ enormously in how automatable they are.

- **Nursing's licence** reserves clinical practice: physical assessment, hands-on care, observation. Those acts are physically irreducible, so the licence protects work that couldn't be automated anyway. The protection compounds.
- **Law's bar card** reserves the courtroom appearance and the signed advice — but not document review, which is why junior law scores 7.9.
- **Pharmacy's licence reserves dispensing and verification — which is precisely the structured, rules-based task a machine performs well.** The protection and the exposure land on the same activity.

That is the crux. Pharmacy's licence is genuinely strong, and it is wrapped around work that technology is genuinely good at. The result is a legal requirement for a human to supervise a process that increasingly does not need supervising.

Two consequences follow.

First, the protection is real but static. A pharmacist must sign off. But if the signing off becomes a formality performed on machine-prepared output, the role thins even as the licence holds. A profession can be legally protected and still hollow out from the inside.

Second, it explains the internal spread precisely. Clinical pharmacy at 4.0 involves judgment on complex regimens and direct patient contact — work the licence protects *and* which resists automation. Central fill at 6.9 is dispensing at scale with minimal patient contact — work the licence protects *and* which automates readily.

The general lesson, and it's the sharpest version of an argument that appears in four profiles in this index: don't ask whether a career has a licence. Ask what the licence reserves, and whether that act is one a machine can perform.

The reversal nobody's careers advice has caught up with

Pharmacy's job market has done something unusual, and a family working from advice more than about three years old will have the wrong picture.

What happened first: a genuine oversupply crisis.

- In 2000 there were **78 accredited pharmacy schools** producing roughly **7,000 graduates a year**
- By 2022 there were **142 schools** and the number of graduates had **nearly doubled to 13,323**
- The National Center for Health Workforce Analysis projected an oversupply of **19,000 to 51,000 pharmacists by 2030**
- Retail chains dominated hiring, and the 2019 National Pharmacist Workforce Study found higher burnout and lower fulfilment among pharmacists at large national chains
- Average pharmacy student debt rose by more than \$40,000 in five years while salaries stagnated

That produced exactly the reputation the profession still carries: expensive degree, saturated market, retail burnout.

What happened next: the pipeline collapsed.

- PharmD applications fell by **more than 60% between 2011 and 2021** — from over 106,000 to under 41,000, per the American Association of Colleges of Pharmacy
- In 2018, **enrolled students exceeded applicants for the first time**
- Enrolment fell from 61,275 in 2012 to 53,516 by 2021
- Graduates declined **24% between 2018 and 2024**
- Mean completions per institution fell from around 109 to around 91

And now the arithmetic has flipped. BLS anticipates **over 14,000 pharmacist openings a year** between 2024 and 2034, while roughly **8,000 PharmD graduates** are expected in 2026 — under 60% of the projected need.

How to read this honestly. Both pictures are partly true, and they describe different settings. Retail dispensing remains under pressure from consolidation, automation and thin margins. Clinical and ambulatory pharmacy is projected to grow around 12% — double the rate for pharmacists overall — and is where the shortage bites.

One caution worth stating. With applications down, some schools have relaxed entry requirements to fill seats. A student should look at a programme's outcomes rather than assuming that admission implies the market needs them.

What the trend shows

Clinical pharmacy: 3.4 → 3.7 → 4.0. Up 0.6.

Central fill: 5.8 → 6.4 → 6.9. Up 1.1.

The gap widened from 2.4 points to 2.9. And the direction of the profession is the thing to watch: pharmacy is moving toward its protected end, with clinical roles growing at roughly twice the rate of the profession overall and technicians absorbing dispensing tasks that pharmacists once performed.

How durable is the protection?

PROTECTION	STRENGTH TODAY	DURABILITY	WHERE THE PRESSURE IS COMING FROM
Regulatory / licensing	Strong	Durable but narrow	Reserves dispensing — the automatable act. Holds, without protecting much.
Human trust / accountability	Real at clinical level	Durable	Someone must answer for a medication error.
Judgment under novelty	Real in clinical practice	Eroding at the dispensing end	Complex regimens resist; routine verification doesn't.
Physical / embodied	Weak	n/a	Verification doesn't require presence — which is why remote dispensing exists.

The honest read. Pharmacy has the licence but not the physical anchor, and that combination is unusual in healthcare. It's why a doctorate-level clinical profession scores in the same band as accounting rather than alongside nursing and dentistry.

How the role will actually change

WAS THE JOB	IS BECOMING THE JOB
Counting and dispensing	Already largely automated
Checking for interactions	Systems flag them; the pharmacist adjudicates
Insurance adjudication	Automated
Verifying routine repeats	Increasingly a formality on machine-prepared output
Managing a complex regimen	Unchanged
Counselling a frightened patient	Unchanged
Catching the error nobody's system flagged	Unchanged

The direction is clear and it's the profession's own strategy: move pharmacists toward clinical services — medication therapy management, chronic disease work, immunisation, prescribing authority where it exists — and away from dispensing.

That transition is the single most important thing happening in pharmacy, and a student should choose a programme on how seriously it prepares them for it.

Human strengths that will matter most

1. Clinical judgment on complex regimens — polypharmacy, interactions, renal dosing, the

patient on fourteen medications. The most protected work in the profession.

2. Catching what the system missed — the error that passes every automated check because it

is contextual rather than rule-based.

3. Counselling — explaining a new diagnosis or a frightening medication to someone who is

scared. Consistently the most valued thing pharmacists do and the least automatable.

4. **Accountability** — being answerable for a medication error. 5. **Working at the clinical interface** — with prescribers, on wards, in ambulatory clinics.

Notably absent: dispensing accuracy, product knowledge recall and throughput. These were the traditional markers of a strong pharmacist and are the fastest-depreciating.

Other things to consider about this job

What's genuinely good about it

The shortage arithmetic is now working in a new graduate's favour — 14,000 projected annual openings against roughly 8,000 graduates is a materially better position than the profession had a decade ago.

Clinical pharmacy is growing at around 12%, roughly double the rate for pharmacists overall, and it's the protected end.

And pharmacy has genuine practical advantages that get overlooked:

- **Regular hours** in most settings, with no routine night work outside hospital rotas
- **High trust** — pharmacists consistently rank among the most trusted professionals
- **Real breadth** — hospital, ambulatory, community, industry, regulatory, academia
- **Clinical work without the training length of medicine** — four years post-degree rather than ten to fifteen
- **Portability** across settings and geographies

What pharmacists report as rewarding: being the accessible healthcare professional — often the only one a patient can see without an appointment; catching errors that would have harmed someone; and the depth of medication expertise, which is genuinely specialist knowledge.

What's genuinely hard about it

The debt is substantial. In 2023, 82.2% of graduating pharmacy students had borrowed to fund their degree, averaging around \$168,000 — up from roughly \$134,000 in 2013, with costs rising faster than inflation since 2004.

Retail conditions are the profession's structural complaint. Chain pharmacy has been associated with higher burnout and lower fulfilment in workforce research, driven by volume

targets, understaffing and expanding non-dispensing duties.

Technicians are absorbing pharmacist tasks, which is efficient and also narrows what the licence practically covers.

And the reputational damage from the oversupply years persists — the profession is still recovering from a decade in which graduates struggled, which is part of why applications fell so far.

Who this work suits — and who it doesn't

Pharmacy tends to suit people who:

- **Are genuinely interested in medicines** rather than in healthcare generally — this is specialist knowledge and it needs to be interesting on its own terms
- **Are precise and comfortable with responsibility for accuracy** where errors harm people
- **Want clinical work without the length of medical training**
- **Are approachable** — being the accessible professional is much of the value
- **Want regular hours** in a clinical field

It's harder for people who:

- Want **physical, hands-on clinical work** — nursing and allied health offer far more of it
- Need **variety** — community dispensing is repetitive by design
- Dislike **retail environments**, which is where much of the employment still sits
- Assumed **any healthcare degree is automatically protected** — that's what this profile is about

What else is moving this market

Scope of practice is expanding, and this is the most consequential trend. Pharmacists in many jurisdictions now immunise, prescribe within protocols, and manage chronic conditions. Every expansion moves the profession toward the protected end, because it substitutes judgment and patient contact for dispensing.

Consolidation and closures in retail pharmacy continue, reshaping where employment sits.

And the pipeline correction is genuine but slow — school capacity adjusts over years, and the shortage projections assume current graduation rates hold.

The AI-native advantage — what to actually do about it

The situation: pharmacy is further into automation than most healthcare, because dispensing was automatable early. That makes the strategic question unusually clear: move toward the clinical end deliberately.

WHAT AN AI-NATIVE PHARMACIST LOOKS LIKE

- 1. Catch what the system missed.** Automated checks find rule-based errors. The errors that harm people are usually contextual — the right drug for the wrong patient situation — and that requires clinical knowledge no system has.
- 2. Work at the clinical interface.** Ward rounds, ambulatory clinics, prescriber collaboration. Every step toward the patient improves the position.
- 3. Take prescribing authority where it exists.** This is the profession's most important structural development, and it converts a dispensing role into a clinical one.
- 4. Specialise.** Oncology, critical care, infectious disease, paediatrics. Specialist clinical pharmacy is the protected end and it's growing fastest.
- 5. Handle patient AI use.** Patients arrive having looked up their medications and side effects, often frightened by what they read. Working with that well is a real and growing skill.

CONCRETE PREPARATION

Before the degree: understand that community dispensing and clinical pharmacy are different careers with different exposure, and choose a programme on its clinical and residency placement record.

During: pursue clinical rotations aggressively. Residency is increasingly what separates clinical roles from dispensing ones, and it's competitive.

And compare programme cost seriously — the debt is substantial and the spread between schools is wide.

The routes in

ROUTE	TYPICAL LENGTH	NOTES
PharmD	4 years post-degree (or 6 direct entry)	The standard route. Compare cost carefully.
PharmD + residency	+1–2 years	Increasingly what separates clinical roles from dispensing. Competitive and worth planning for early.
Clinical specialisation	+1–2 years beyond residency	Oncology, critical care, ID. The most protected work.
Pharmacy technician	Under 1–2 years	Growing around 7%, absorbing dispensing tasks. Short route, lower ceiling.
Pharmaceutical industry / regulatory	Post-qualification	Uses the knowledge without patient contact — and scores accordingly.

Where a pharmacy qualification can lead later

Hospital and clinical pharmacy, ambulatory and specialty practice, community pharmacy and ownership, pharmaceutical industry, regulatory affairs, medical science liaison roles, academia and research, and health policy.

The pattern in this index holds: roles with patient contact and clinical accountability score better than roles processing medication. Industry and regulatory work uses the expertise without the protection.

What to look for in a pharmacy program

- 1. Clinical rotation depth and variety.** How much hospital and ambulatory exposure, not just community placement?
- 2. Residency match rate.** This is the number that increasingly determines whether a graduate reaches clinical practice.
- 3. Total cost — and the debt-to-income ratio.** Average borrowing around \$168,000 against a profession with historically stagnant salaries deserves careful arithmetic.

4. Does the curriculum reflect expanded scope? Prescribing, immunisation, chronic disease management — the direction the profession is moving.

5. Outcomes by setting. What proportion of graduates are in clinical roles versus retail dispensing at three years?

Questions to ask a pharmacy program

On outcomes:

- What proportion of your graduates match into residency?
- Where are graduates working at three years — hospital, ambulatory, community, industry?
- How has that mix changed over five years?

On cost:

- What's the total cost of attendance, and the median debt at graduation?

On the curriculum:

- How much clinical and ambulatory rotation time, and in what settings?
- Does the curriculum prepare students for prescribing authority and expanded scope?
- How does it address automated dispensing and AI-assisted verification?

On admissions — ask this one directly:

- Have your entry requirements changed in the last five years? With applications down more than 60% from their peak, this is a fair question and the answer is informative.

Red flags: low residency match rates; predominantly community placements; no engagement with expanded scope; evasiveness about graduate destinations or admissions standards.

Go deeper — further reading

- **Pharmacists — Occupational Outlook Handbook** — US Bureau of Labor Statistics. Projected openings, median pay and entry requirements. Free, and the source of the

14,000 annual openings figure. Read the **Pharmacy Technicians** page alongside it.

- **AACP — pharmacy school application and enrolment data** — American Association of Colleges of Pharmacy. The primary source for the 60% application decline and enrolment figures. The clearest picture of the pipeline reversal described in section 5.
- **Insights into decreasing applicants to schools and colleges of pharmacy** — University of Washington Center for Health Workforce Studies. Peer-reviewed analysis of why applications fell, including the debt figures. Academic rather than promotional.

The counterargument — read this one:

There are two, pulling in opposite directions, and a family should weigh both.

Our score is too high: the pipeline has collapsed, BLS projects 14,000 annual openings against roughly 8,000 graduates, and clinical pharmacy is growing at 12%. On this reading, a profession heading into shortage with expanding scope of practice is far better positioned than a 4.9 suggests, and we are scoring the automatability of a task the profession is actively moving away from.

Our score is too low: the licence reserves dispensing, dispensing is automating, and technicians are absorbing what remains. A profession can hold its legal protection while the work inside it thins — and the scope expansion is a strategy, not yet an outcome.

Where we land: we hold the score because it reflects the work as currently constituted, and most pharmacists still spend most of their time on dispensing-adjacent activity. But we think the optimistic case is stronger here than in most profiles, and the specific thing we're watching is the proportion of pharmacists in clinical and ambulatory roles. If that share rises materially, our score should come down and we'll say so.

Bottom line

Pharmacy is a good career being measured at an awkward moment in its own transition.

Its score is the weakest in licensed healthcare for a specific and instructive reason: pharmacy's licence reserves dispensing, and dispensing is exactly what machines do well. A licence is only as protective as the act it reserves. Nursing's protects work that couldn't be automated anyway; pharmacy's protects work that could.

But the profession is moving, and the job market has already turned. Applications fell more than 60% from their peak and graduates declined 24% between 2018 and 2024 — so a profession that produced an oversupply crisis is now projected to graduate under 60% of what BLS expects to be needed. Clinical pharmacy is growing at roughly twice the rate of the profession overall.

So the useful advice is unusually clear: aim at clinical and ambulatory practice deliberately rather than assuming a PharmD leads there. Plan for residency early, because it increasingly separates clinical roles from dispensing ones. Pursue prescribing authority and expanded scope wherever it exists. Compare programme cost carefully against borrowing that now averages around \$168,000. And ask any school directly what proportion of its graduates are in clinical roles at three years.

And one honest thing worth saying. Pharmacy offers genuine clinical work, high public trust and regular hours, four years after a bachelor's rather than ten to fifteen. That combination is rare — and it's available to someone who aims at the right end of the profession from the start.

Discussion questions

Part 1 — About the work itself

1. What made pharmacy appealing? Push past "healthcare" to something specific. 2. **Are they interested in medicines specifically**, or in healthcare generally? This is

specialist knowledge and it needs to be interesting on its own terms.

3. Do they want **hands-on clinical work**? Because nursing and allied health offer far more of

it, at better scores.

4. What does a Tuesday look like — in a hospital, and in a community pharmacy? They're very

different jobs.

Part 2 — Reacting to the findings

5. Pharmacy scores worse than nursing, dentistry and physical therapy despite being licensed and

doctorate-level. Does the reason in section 4 make sense to them?

6. **A licence is only as protective as the act it reserves.** Can they apply that idea to

another career they're considering?

7. Clinical pharmacy scores 4.0 and central fill 6.9. Does that change where they'd aim?

Part 2b — The harder conversation

8. **The debt.** Around \$168,000 average borrowing. Have you run that against realistic starting

salaries?

9. Do they understand that residency increasingly determines whether they reach clinical practice

rather than dispensing?

10. Have you asked schools what proportion of graduates are in clinical roles at three years?

Part 2c — The other side

11. **The pipeline has reversed.** Applications down over 60%, graduates down 24%, and BLS

projecting more openings than graduates. Does that change the picture?

12. Of what pharmacists find rewarding — accessibility to patients, catching errors, medication

expertise, regular hours — which two matter most?

13. Looking at "who this work suits": which items sound like them? Answer separately, then compare.

Part 2d — The AI question, practically

14. Automated systems catch rule-based errors; the ones that harm people are usually contextual.

What does that suggest about where the value sits?

15. Scope of practice is expanding — prescribing, immunisation, chronic disease management. Is

that the version of the job they want?

Part 3 — Testing it against reality

16. **The most valuable single action:** shadow in both a hospital pharmacy and a busy community

one. The gap between those two days is essentially this whole document.

17. Find a pharmacist three to five years qualified and ask what proportion of their time is

dispensing versus clinical.

Part 4 — About the program

18. Ask each programme the residency match and graduate-destination questions from section 14. 19. **Have their entry requirements changed in the last five years?** With applications down

sharply, that's a fair and informative question.

Part 5 — Widening the frame

20. Have they compared this properly with **nursing**? Scores 2.8, reaches licensed practice

faster, and costs considerably less.

21. What about **medicine**? Longer and more expensive, but a materially better score and

broader scope.

22. If pharmacy weren't available, what would they choose? The answer usually reveals what

they're actually drawn to.

Part 6 — For the parent, alone

23. Did I assume healthcare degrees are automatically protected? Has this changed that? 24. Am I comfortable with around \$168,000 of borrowing, and have I said so either way? 25. What's the one thing I most want them to understand — in a sentence, without a number in it?

This is analysis and informed judgment about a fast-moving situation. For pharmacy specifically, we've flagged that the profession is actively transitioning toward clinical practice, that our score reflects the work as currently constituted rather than where it is heading, and that the job market has reversed in a way most careers advice has not caught up with.

Scores for 2023 and 2025 are retrospectively calculated using the current methodology. Scores measure exposure to what AI can already do — not how much any particular employer has deployed.

Technical scoring appendix — Pharmacy

Pivotum Degree Risk Index — Fall 2026 Edition

This appendix is the audit trail. It sets out the formula, the rating anchors, every factor rating behind every track score in this profile, the backcast arithmetic, a sensitivity analysis, and the specific things that would change our mind.

We publish it because a score nobody can check is an opinion with a decimal point.

1. The formula

$$\begin{aligned}\text{Risk} = & 0.35 \times \text{Automatability} \\ & + 0.15 \times \text{EntryErosion} \\ & + 0.15 \times (10 - \text{Physical}) \\ & + 0.15 \times (10 - \text{Trust}) \\ & + 0.10 \times (10 - \text{Regulatory}) \\ & + 0.10 \times (10 - \text{Judgment})\end{aligned}$$

Two exposure factors carry 50% of the weight; three protection factors carry 40%; judgment under novelty carries the remaining 10% and sits on the protection side. The deliberate consequence is that protection can outweigh exposure — which is why a highly automatable job like teaching still scores low.

Bands. Very low below 3.0 · Low 3.0–4.4 · Moderate 4.5–5.4 · Moderate–High 5.5–6.9 · High 7.0 and above.

2. Rating anchors

HOW THE 0–10 RAW RATINGS ARE ANCHORED

Every factor is rated on its own 0–10 scale before weighting. The anchors are identical across all twenty-seven careers, which is what makes the scores comparable.

Task automatability (A) — what share of the working week could a capable current system already do to an acceptable standard?

RATING	ANCHOR
--------	--------

- | | |
|------|---|
| 0–2 | Almost nothing. The work is physical, relational or wholly novel. |
| 3–4 | Administrative surround only — notes, scheduling, correspondence. |
| 5–6 | A meaningful minority of the week, mostly documentation and lookup. |
| 7–8 | A large share, including tasks central to the junior version of the role. |
| 9–10 | Most of the described duties, at or near professional standard. |

Entry-path erosion (E) — *how much has the traditional route into this work narrowed?*

RATING	ANCHOR
--------	--------

- | | |
|------|---|
| 0–2 | Protected by statute or physical necessity. Training hours cannot be automated. |
| 3–4 | Stable. Some junior tasks absorbed; the apprenticeship survives. |
| 5–6 | Visible narrowing. Employers prefer experience; junior intake reduced. |
| 7–8 | Substantial. The work juniors learned on has largely gone. |
| 9–10 | The training layer and the automatable layer were the same layer. |

Physical / embodied (P) — *must this be done in person, with a body, in an unpredictable setting?*

RATING	ANCHOR
--------	--------

- | | |
|------|--|
| 0–2 | Fully screen-based. Location-independent. |
| 3–4 | Occasional presence; the core work is not physical. |
| 5–6 | Regular physical presence, in largely controlled conditions. |
| 7–8 | Substantially physical, with some variability of setting. |
| 9–10 | Irreducibly physical, in conditions that differ every time. |

Human trust / accountability (T) — *does someone need a specific human they can rely on, and must a human answer when it goes wrong?*

RATING	ANCHOR
0–2	Output is anonymous and reviewed by others.
3–4	Work is signed off by someone else. No personal reliance.
5–6	Some client relationship; accountability is institutional.
7–8	Named responsibility, with commercial or professional consequence.
9–10	The relationship is the service, and liability attaches personally.

Regulatory / licensing (R) — *does the law reserve this act to a qualified human?*

RATING	ANCHOR
0–2	No licence exists and none is proposed.
3–4	Certification is customary but not reserved.
5–6	Partial reservation, varying by jurisdiction or task.
7–8	Licensed practice, with some unreserved adjacent work.
9–10	Comprehensive statutory reservation, including of the training route.

Judgment under novelty (J) — *how often does the work present something with no matching precedent, where being wrong is costly?*

RATING	ANCHOR
0–2	Routine by design. Deviation is an error, not a case.
3–4	Occasional exceptions, handled by escalation.
5–6	Regular non-standard cases within a known range.
7–8	Frequent genuine novelty with material consequences.
9–10	Novelty is constant, and in some fields adversarially generated.

3. Factor ratings by track

Ratings are the Fall 2026 edition unless a range is shown. **P, T, R and J are held constant across editions** — those protections are structural and do not move on a three-year horizon. All measured movement is in A and E.

CLINICAL / HOSPITAL PHARMACY — 4.0 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.0	5.7	6.4	2.24
Entry-path erosion	15%	2.5	2.8	3.2	0.48
Physical / embodied	15%	5.0	5.0	5.0	0.75
Human trust / accountability	15%	8.5	8.5	8.5	0.22
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	7.5	7.5	7.5	0.25
				Total	3.99 → 4.0

AMBULATORY & SPECIALTY PHARMACY — 4.4 (LOW)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.5	6.2	6.9	2.42
Entry-path erosion	15%	2.8	3.2	3.5	0.53
Physical / embodied	15%	4.5	4.5	4.5	0.82
Human trust / accountability	15%	8.2	8.2	8.2	0.27
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	7.0	7.0	7.0	0.3
				Total	4.38 → 4.4

COMMUNITY / RETAIL PHARMACY — 4.9 (MODERATE)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.3	6.3	7.0	2.45
Entry-path erosion	15%	3.2	3.6	4.0	0.6
Physical / embodied	15%	3.5	3.5	3.5	0.97
Human trust / accountability	15%	7.5	7.5	7.5	0.38
Regulatory / licensing	10%	9.5	9.5	9.5	0.05
Judgment under novelty	10%	5.5	5.5	5.5	0.45
				Total	4.9 → 4.9

PHARMACY TECHNICIAN — 5.7 (MODERATE-HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	4.6	5.8	6.7	2.34
Entry-path erosion	15%	4.0	4.5	5.0	0.75
Physical / embodied	15%	3.5	3.5	3.5	0.97
Human trust / accountability	15%	5.5	5.5	5.5	0.67
Regulatory / licensing	10%	6.5	6.5	6.5	0.35
Judgment under novelty	10%	4.0	4.0	4.0	0.6
				Total	5.69 → 5.7

CENTRAL FILL / MAIL ORDER — 6.9 (MODERATE-HIGH)

FACTOR	WEIGHT	2023	2025	2026	CONTRIBUTION TO 2026 SCORE
Task automatability	35%	5.8	7.2	8.4	2.94
Entry-path erosion	15%	5.5	6.2	6.8	1.02
Physical / embodied	15%	2.0	2.0	2.0	1.2
Human trust / accountability	15%	4.0	4.0	4.0	0.9
Regulatory / licensing	10%	8.5	8.5	8.5	0.15
Judgment under novelty	10%	3.0	3.0	3.0	0.7
				Total	6.91 → 6.9

4. Backcast arithmetic

Worked example for the headline track, showing exactly how the three-year movement is composed.

Clinical / hospital pharmacy

EDITION	A	E	PROTECTION DEFICIT (FIXED)	SCORE
2023	5.0	2.5	1.27	3.4
2025	5.7	2.8	1.27	3.7
Fall 2026	6.4	3.2	1.27	4.0

Movement decomposition: automatability contributed +0.49 and entry-path erosion +0.11, for a total of +0.6 points over three years.

The 2023 and 2025 figures are retrospective reconstructions using the current methodology, not archived scores from published editions. They are our best assessment of what each factor would have rated at the time, given what was then demonstrable. They are not measurements.

5. Sensitivity

How far the score moves if a single factor is rated one point differently:

FACTOR	±1 POINT MOVES THE SCORE BY
Task automatability	±0.35
Entry-path erosion	±0.15
Physical / embodied	±0.15
Human trust / accountability	±0.15
Regulatory / licensing	±0.10
Judgment under novelty	±0.10

What this means in practice. A one-point disagreement about automatability moves a score by 0.35 — enough to shift a track between bands at the margins. A one-point disagreement about licensing moves it by 0.10, which almost never changes a band.

So if you disagree with one of our numbers, the automatability rating is where the disagreement matters most, and it is the rating we would most want challenged.

6. Stated limitations

We measure capability, not deployment. Scores reflect exposure to what AI can already do at the leading edge — not how much any particular employer has adopted. Adoption lags capability by years and lags unevenly: large, well-funded organisations first; small, rural, public-sector and thin-margin employers much later. The lag is a buffer, not a shelter. It buys time without changing direction.

We do not measure demand. A task can be highly automatable and highly in demand simultaneously. Where the labour market and our framework disagree, we say so in the profile rather than adjusting the score to fit.

We do not measure oversupply. The number of people entering a field is outside our six factors, and for some careers it matters more than automation.

We do not measure industry economics. A profession can contract for reasons entirely unrelated to technology.

We do not measure whether the work is worth doing. Pay, satisfaction, debt, hours and meaning sit outside the score and are covered separately in each profile — often at greater length, because for several careers they matter more.

Sub-track definitions are ours. Job titles vary between employers and countries. We have chosen tracks that reflect genuinely different work, not official classifications.

7. What would change our mind

Each edition names specific, checkable indicators. If these move, the scores move — including downward.

Across the index:

- **Graduate hiring share.** If employers in the professions where we rate entry-path erosion highly return to hiring juniors at pre-2023 rates, those ratings fall.
- **Content of mandated training hours.** Several professions protect their on-ramp through required supervised experience. If the content of those hours shifts from producing work to supervising it, the protection weakens without any rule changing.
- **Offsite and prefabricated construction share.** The clearest test of our physical-protection ratings: automation performs well in controlled settings, so moving work indoors raises exposure without any robotics breakthrough.
- **Verified deployment rather than capability.** If adoption stalls materially below capability for several editions, our leading-indicator approach is overstating near-term risk and we will say so.

We will report movement in either direction. A framework that only ever revises upward is not measuring anything.

Scores measure exposure to what AI can already do — not how much any particular employer has deployed. The 2023 and 2025 figures are retrospective reconstructions using the current methodology.